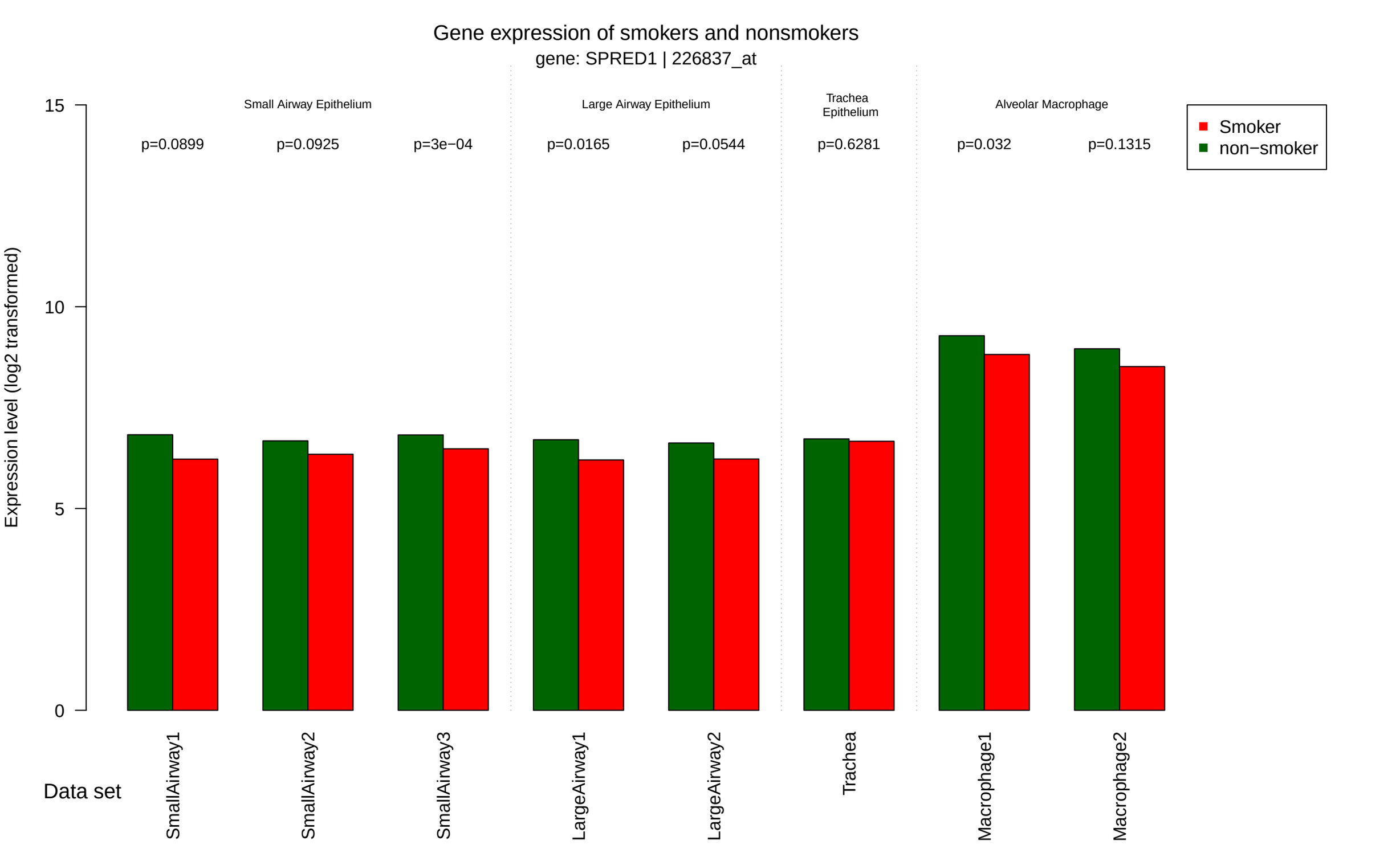
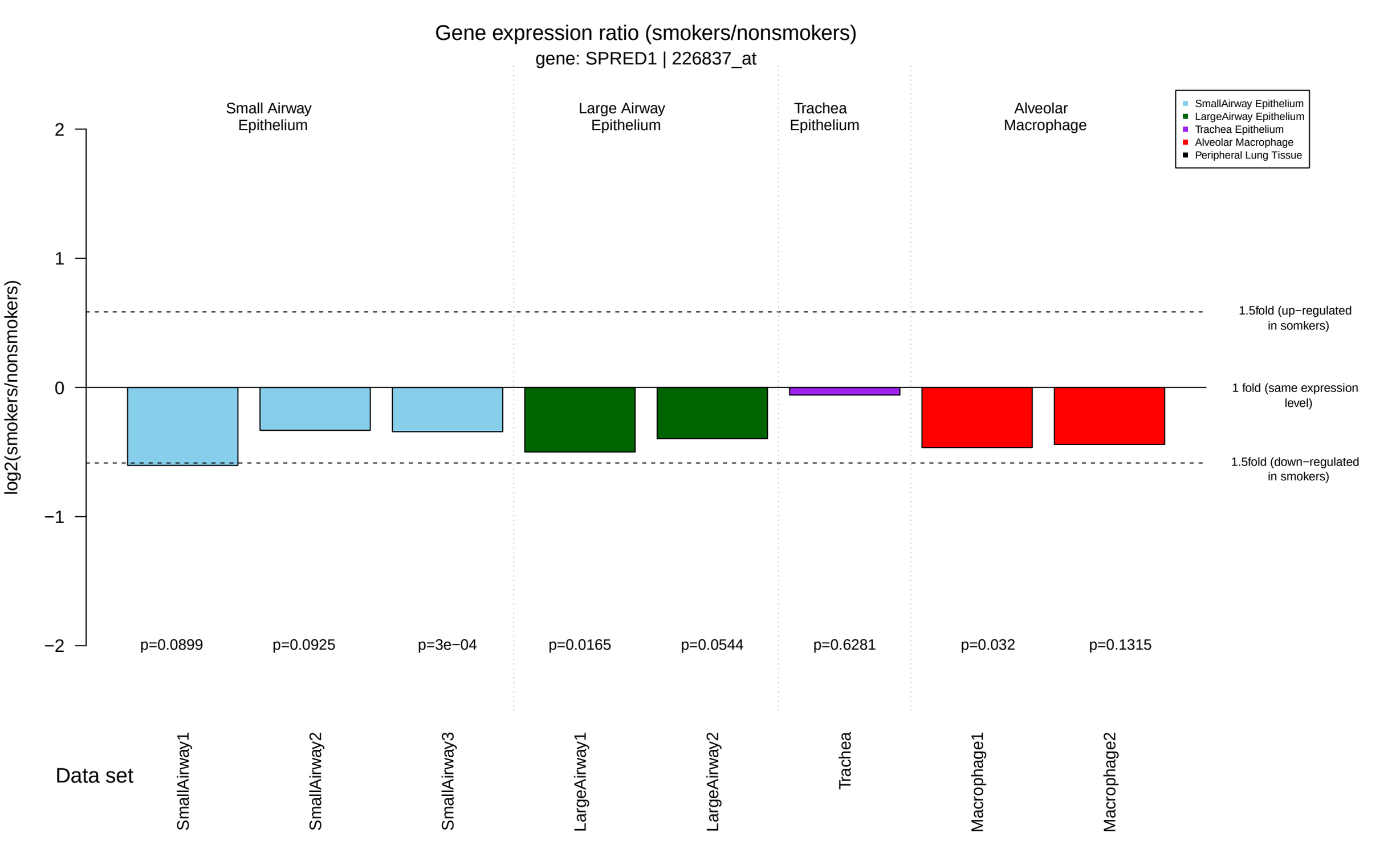
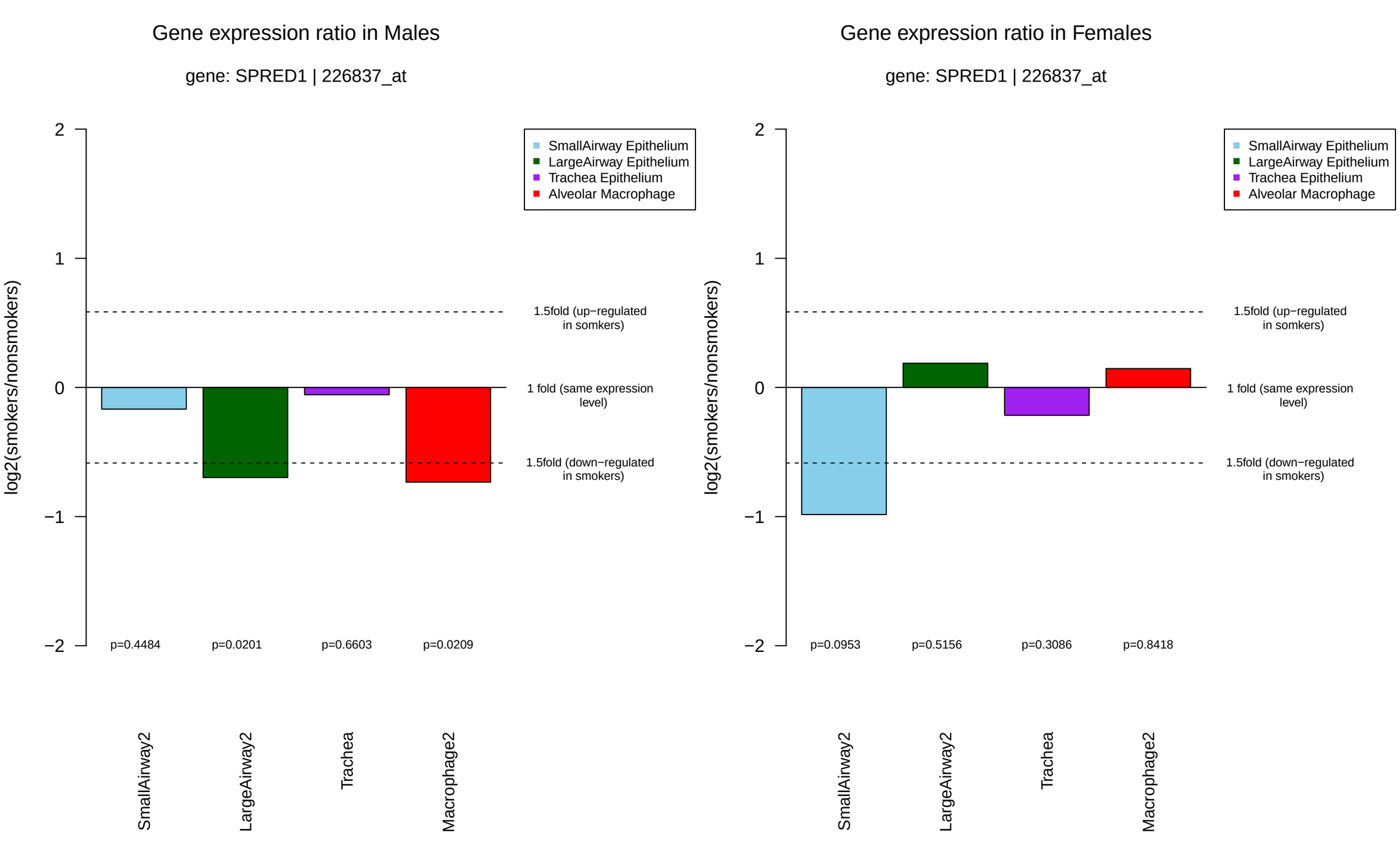
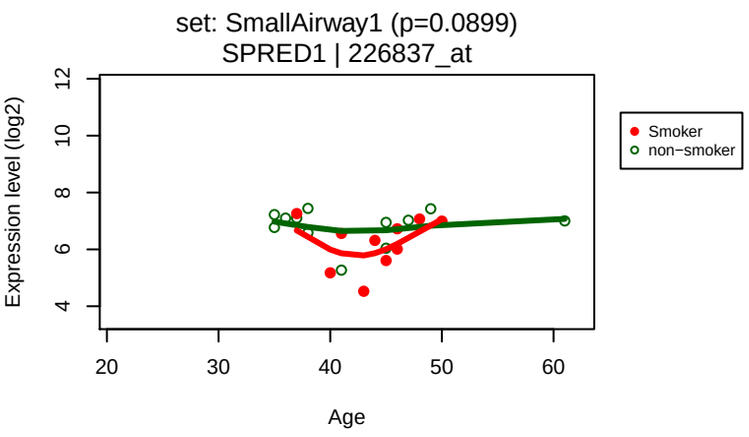
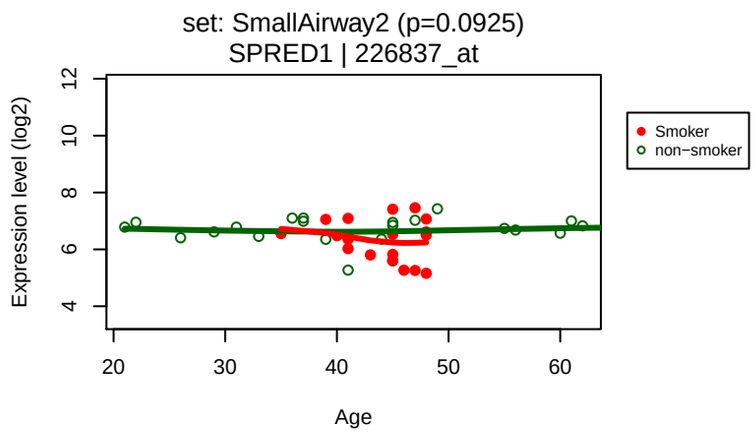
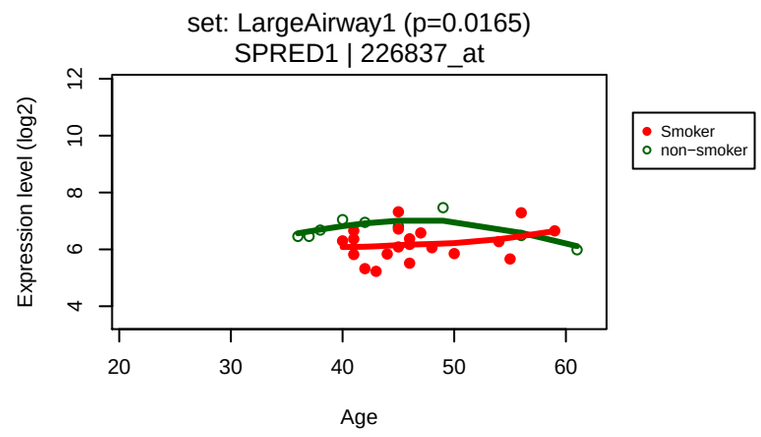

Gene expression of smokers and nonsmokers
gene: SPRED1 | 226837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0899
p=0.0925
p=3e−04
p=0.0165
p=0.0544
p=0.6281
p=0.032
p=0.1315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

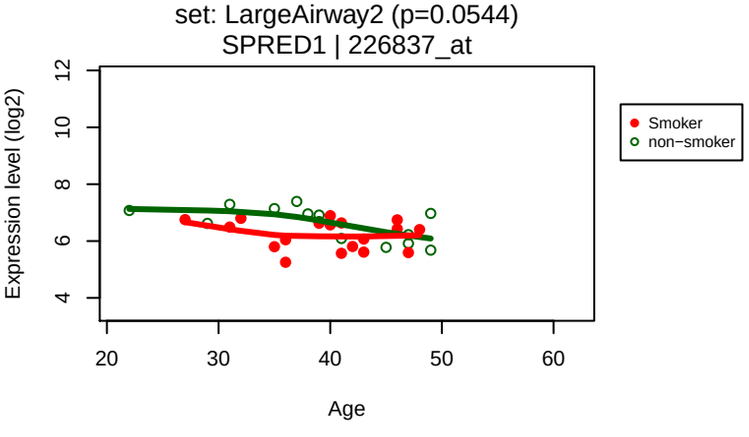

Gene expression ratio (smokers/nonsmokers)
gene: SPRED1 | 226837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0899
p=0.0925
p=3e−04
p=0.0165
p=0.0544
p=0.6281
p=0.032
p=0.1315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

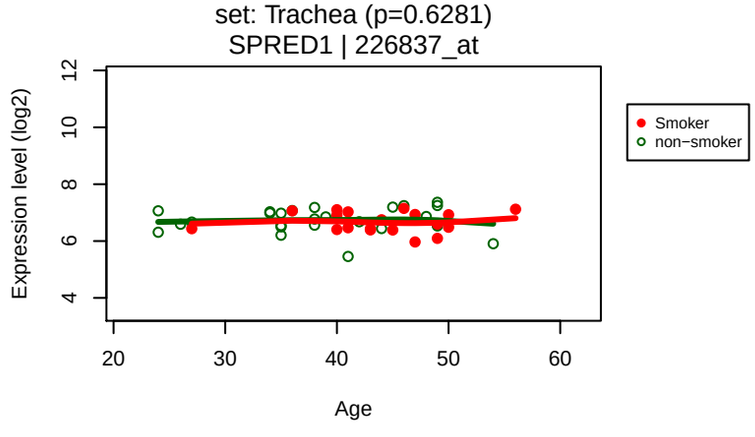

Gene expression ratio in Males
gene: SPRED1 | 226837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4484
p=0.0201
p=0.6603
p=0.0209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPRED1 | 226837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0953
p=0.5156
p=0.3086
p=0.8418
SmallAirway2
LargeAirway2
Trachea
Macrophage2

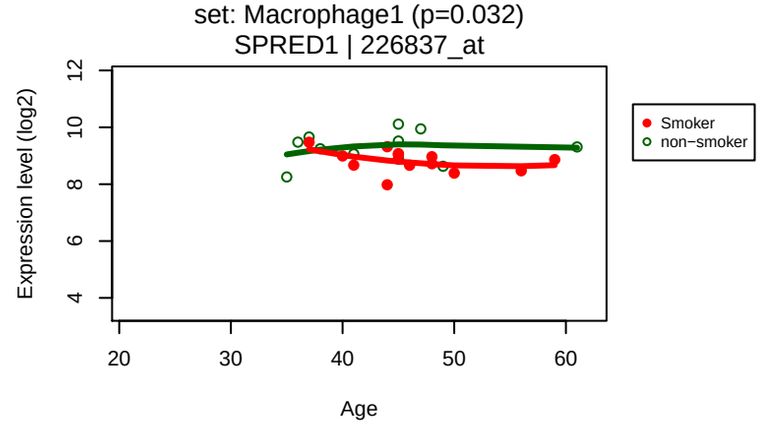

set: SmallAirway1 (p=0.0899)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker

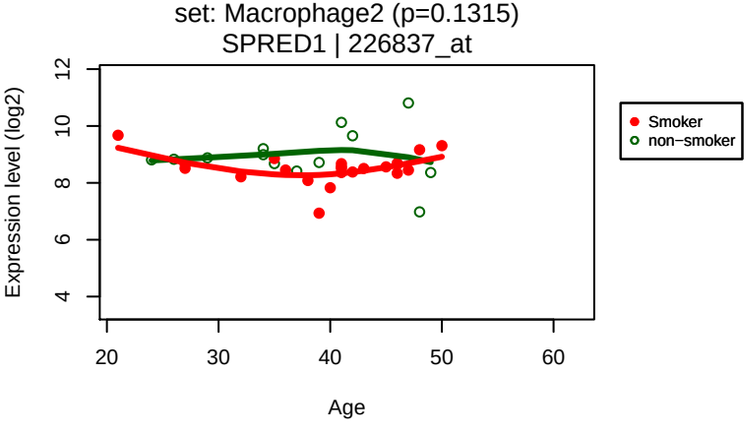

set: SmallAirway2 (p=0.0925)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker

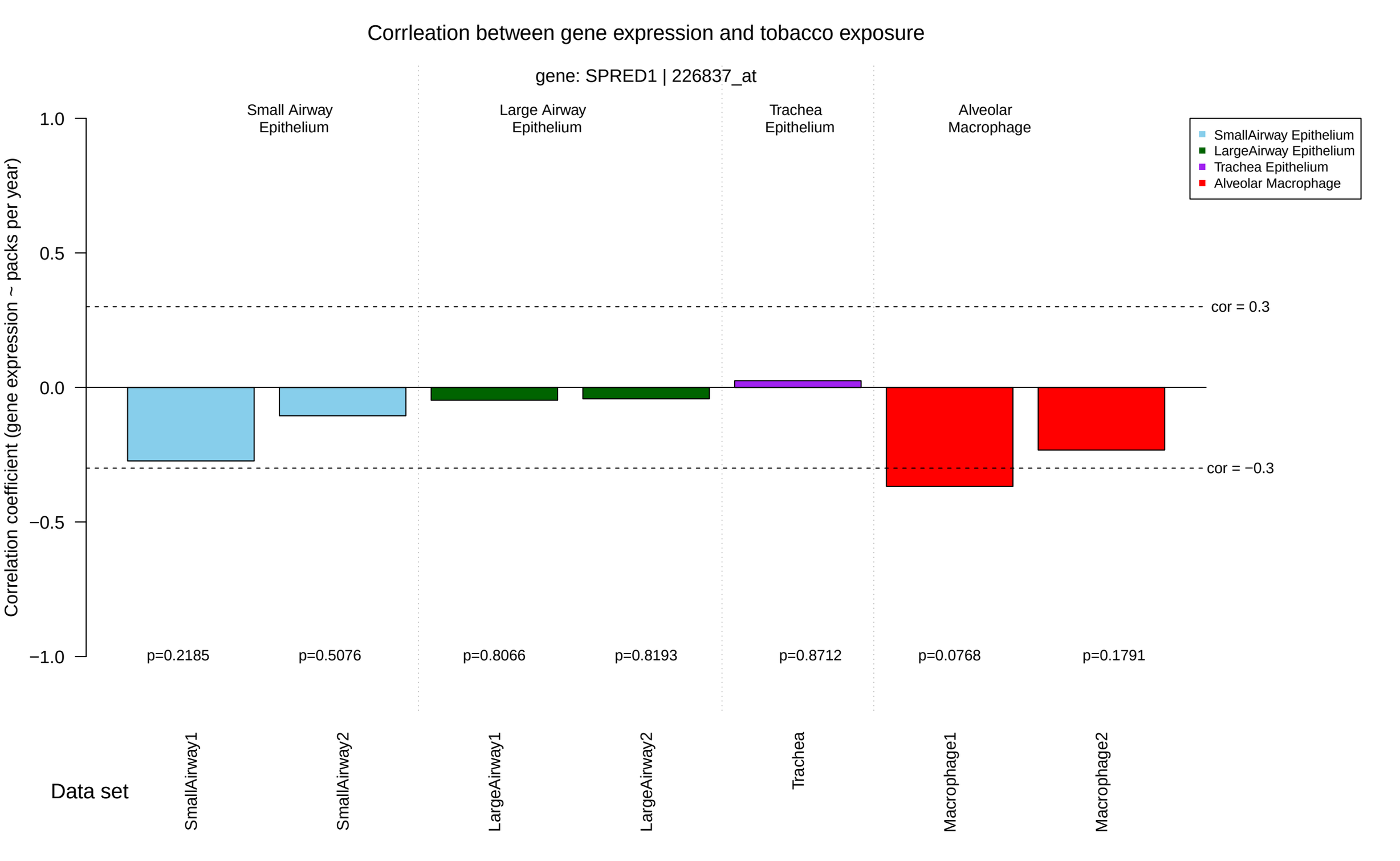

set: LargeAirway1 (p=0.0165)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0544)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6281)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.032)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1315)
SPRED1 | 226837_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPRED1 | 226837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2185
p=0.5076
p=0.8066
p=0.8193
p=0.8712
p=0.0768
p=0.1791
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2